

Compact Abelian Torsion Groups Have Bounded Exponent

Candidate full answer for arXiv:1306.1480

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Status. Candidate full proof, likely valid, pending specialist human review. Under the standard meaning of “torsion group”, the bounded-exponent hypothesis in the source theorem is automatic.

1 Source question

Czuron and Wojciechowski prove a quantitative nonisomorphism theorem for a “commutative compact torsion group with finite exponent” and then ask whether the assumption of bounded exponent is significant [1, p. 3].

$\gcd(x_1, x_2, \dots, x_{n+1}) = 1$, and there is no vacuous sums $x_{i_1} + x_{i_2} + \dots + x_{i_m}$, has the number of integral solutions $\mathbf{x} = (x_1, x_2, \dots, x_{n+1})$ bounded by $C_1 \cdot \exp(C_2 n^3 \log n)$ where the constants C_1 and C_2 depend on M only.

Remark. We do not know whether the assumption of bounded exponent is significant. Evidently the method used in our proof does not work in that case. However we do not know any counterexample neither.

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Figure 1: The open remark in arXiv:1306.1480, PDF page 3.

For the class literally stated in the theorem, the answer is no: compactness, commutativity, and torsion already force one common exponent.

2 The topological group lemma

We write compact groups additively. A group is *torsion* if every one of its elements has finite order, and it has *finite exponent* if some positive integer annihilates every element.

Lemma 1. *Every compact Hausdorff abelian torsion group has finite exponent.*

Proof. Let G be such a group. For each $n \geq 1$, set

$$K_n = \ker(n \operatorname{id}_G) = \{x \in G : nx = 0\}.$$

Because multiplication by n is a continuous homomorphism, K_n is a closed subgroup of G . The torsion assumption gives the countable cover

$$G = \bigcup_{n=1}^{\infty} K_n.$$

Every compact Hausdorff space is a Baire space. Hence some K_N has nonempty interior. A subgroup with nonempty interior is open: if U is a nonempty open subset of K_N and $u \in U$, then $U - u$ is a neighbourhood of 0 contained in K_N , and all translates of that neighbourhood show that K_N is open.

The quotient G/K_N is therefore both compact and discrete, so it is finite. Let q be its exponent (one may simply take $q = |G/K_N|$). For every $x \in G$, the coset $x + K_N$ is killed by q , which says $qx \in K_N$. Applying N now gives

$$Nq, x = 0.$$

Thus the single integer Nq annihilates all of G . □

Corollary 1. *In the main theorem of arXiv:1306.1480, the phrase “with finite exponent” can be deleted without changing the class of groups. In particular, for every compact abelian torsion group G satisfying the paper’s subgroup exclusion, all of the theorem’s distortion conclusions and quantitative lower bounds hold exactly as stated.*

Proof. Apply Lemma 1 to G , then invoke the source theorem. □

3 Why the hypotheses matter

The Baire step is the entire point. Compactness cannot simply be dropped: the discrete abelian group

$$\bigoplus_{m=2}^{\infty} \mathbb{Z}/m\mathbb{Z}$$

is torsion but has no finite exponent. Commutativity is used to ensure that K_n is a subgroup; for the source theorem the group is explicitly commutative.

There is also an important terminological boundary. A profinite or zero-dimensional compact group need not be torsion: for example, $\prod_{m \geq 2} \mathbb{Z}/m\mathbb{Z}$ contains elements of infinite order. Thus the argument resolves the question under the standard algebraic meaning of the words appearing in the source theorem. If “torsion” was intended nonstandardly to mean merely profinite, or if the intended object was instead an unbounded-exponent discrete dual group, that would be a different problem not formulated by the displayed theorem.

4 Verification and review recommendation

The proof uses only four exact facts:

1. the torsion condition gives $G = \bigcup_n K_n$;
2. each K_n is a closed subgroup;
3. Baire category makes one K_N open;
4. a compact discrete quotient is finite.

No quantitative estimate, classification theorem, or unproved regularity claim is needed.

Review as a likely valid full answer to the literal question. The only substantive review point is semantic: confirm that “compact torsion group” is being used in its standard sense (every element has finite order). Under that reading, the finite-exponent assumption is provably redundant.

References

- [1] A. Czuron and M. Wojciechowski, *On the isomorphisms of Fourier algebras of finite abelian groups*, arXiv:1306.1480, 2013. <https://arxiv.org/abs/1306.1480>.